

Microstructure and wear behavior of $\text{Al}_x\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_y$ high-entropy alloys

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Abstract

A series of $\text{Al}_x\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_y$ high-entropy alloys with different Al and Ti contents were designed, and their phase and microstructure were investigated. The adhesive wear behavior and mechanism were also studied and compared with the conventional wear-resistant steels SUJ2 and SKH51. The amounts of Al and Ti strongly affect the phase and microstructure, particularly the amount and morphology of the hard $\eta\text{-(Ni, Co)}_3\text{Ti}$ phase. It was found that the wear resistance of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}$ and $\text{Al}_{0.2}\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}$ alloys is at least two times better than that of conventional wear-resistant steels with similar hardness. The excellent anti-oxidation property and resistance to thermal softening in these high-entropy alloys are proposed to be the main reasons for the outstanding wear resistance. © 2011 Acta Materialia Inc. Published by Elsevier Ltd. All rights reserved.

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1. Introduction

Most metallic materials, such as steel and aluminum alloys, are based on one or two major elements. Various alloying elements are added to the materials to improve their properties. Differing from conventional alloy design, Yeh et al. [1,2] proposed a new alloy design concept in 1995 in which more than five major elements, each having an atomic percentage (at.%) in the range 5–35 [2–4], are combined. This new approach to alloy design with multiple principal elements leads to the formation of “high-entropy alloys” (HEAs), in which certain effects not occurring in conventional alloys are observed; these include high-entropy, sluggish-diffusion, lattice-distortion and cocktail effects [5]. Among these core effects, the high-entropy effect is the most important (hence the name). The high-entropy effect originates from the significantly higher configura-

tional entropy in HEAs compared to that in conventional alloys. Such high mixing entropy can favor the formation of disordered solution states and suppress the formation of intermetallic compounds, which is of paramount importance in the application of these alloys [2,6].

Previous studies [2,3,5–18] have shown that HEAs have many attractive properties, such as high hardness, good oxidation resistance, high elevated-temperature strength and good age-softening resistance. However, the cost of HEAs may be higher than that of conventional alloys (such as steel and Al alloys) because of the use of multiple elements; though it is still cheaper than most superalloys and Ti alloys. Thus, HEAs show great potential for use in tools, molds and structural components.

Previous research has indicated that $\text{Al}_x\text{CoCrCuFeNi}$ [9,12,19], $\text{Al}_{0.5}\text{CoCrCuFeNiTi}_x$ [11] and AlCoCrFeNiTi_x [13] HEAs have a variety of microstructures and mechanical properties. These alloys have body-centered cubic (bcc) or face-centered cubic (fcc) crystal structures, or the mixture of both. In general, the bcc-based alloys have higher strength but lower workability than the fcc-based alloys

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and vice versa [6,9]. Based on these facts, a $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$ HEA having an fcc crystal structure was designed. In this alloy, the molar ratios of strong fcc formers, Co and Ni, are increased to 1.5 to ensure that the fcc structure is retained as the primary phase [20]. This renders the alloy workable by plastic deformation, while remaining quite strong and tough. In addition, the strength can be enhanced by precipitation hardening like in Ni-based superalloys. In this study, four $\text{Al}_x\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_y$ alloys, with different Al and Ti contents, based on the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$, were designed, and the effects of Al and Ti concentrations on the phase, microstructure, precipitation and mechanical properties were investigated.

Wear is an important phenomenon in many applications. However, information on the wear behavior of HEAs is very limited, and most reported studies have been conducted under abrasive conditions [7,10–12,21]. In this study, we performed adhesive wear tests on $\text{Al}_x\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_y$ alloys to study the wear mechanism and compare the results with the conventional wear-resistant steels SUJ2 and SKH51.

2. Experimental procedure

Four alloys with different Al and Ti contents were studied: $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$ ($\text{Co}_{27.3}\text{Cr}_{18.2}\text{Fe}_{18.2}\text{Ni}_{27.3}\text{Ti}_{9.1}$), $\text{Al}_{0.2}\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$ ($\text{Al}_{3.5}\text{Co}_{26.3}\text{Cr}_{17.5}\text{Fe}_{17.5}\text{Ni}_{26.3}\text{Ti}_{8.8}$), $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}$ ($\text{Co}_{25}\text{Cr}_{16.7}\text{Fe}_{16.7}\text{Ni}_{25}\text{Ti}_{16.7}$) and $\text{Al}_{0.2}\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}$ ($\text{Al}_{3.2}\text{Co}_{24.2}\text{Cr}_{16.1}\text{Fe}_{16.1}\text{Ni}_{24.2}\text{Ti}_{16.1}$); these alloys are henceforth referred to as Al00Ti05, Al02Ti05, Al00Ti10 and Al02Ti10, respectively. All alloy ingots were prepared by arc-melting the mixtures of the constituent elements with purities higher than 99.5 wt.% in an argon atmosphere, and the melting procedure was repeated several times to improve the homogeneity of the ingots. As-cast alloys were homogenized at 1100 °C for 4 h and air cooled. The alloys were then aged at 800 °C for 10 h followed by air cooling. Both homogenization and aging treatments were conducted in air.

For microstructure investigations, the specimens were ground and polished using standard metallographic techniques, and to reveal grain boundaries and fine precipitates, some specimens were etched with dilute aqua regia. The wear behavior of the alloys was investigated by pin-on-disk sliding tests based on the ASTM G99 standard under dry sliding conditions. SUJ2 (AISI 52100) bearing steel and SKH51 (AISI M2) high-speed steel were selected as reference materials in the wear test because they have good mechanical properties and are proven anti-wear materials. The reference materials and HEAs were tested under identical conditions. The wear test specimens were cylindrical in shape, and each had a diameter of 8 mm and height of 3 mm. Each specimen was fixed in a holder to wear against a 75 mm diameter SKH51 (AISI M2) steel disk, which had a hardness value of HV 890. The distance between the specimen and the center of the disk was 20 mm. The sliding speed was 0.5 m s^{-1} , with a normal

load of 29.4 N. To obtain sufficient wear loss, the required sliding distance for harder alloys was greater: the sliding distance for the softer alloys (Al00Ti05 and Al02Ti05) was 5400 m, while that for the harder alloys (Al00Ti10, Al02Ti10, SUJ2 and SKH51) was 43,200 m. The anti-wear property is measured by wear resistance, which is defined as $W_r = L/\Delta V$, where W_r is the wear resistance and ΔV is the volumetric loss of the specimen after it has slid through a distance L , and is obtained by dividing the weight loss by the sample density. Crystal structures were identified using an X-ray diffractometer (Rigaku ME510-FM2) under radiation conditions of 30 kV, 20 mA, a Cu target and a scanning speed of 2° min^{-1} . The microstructure and chemical composition of the alloys, the wear surface and wear debris were analyzed using a scanning electron microscope (SEM; JEOL JSM-5410) with an attached X-ray energy dispersive spectrometer (EDS). Room-temperature macro and microhardness measurements were performed using Mitutoyo HV-115 and HM-115 hardness testing machines with loads of 5 and 25 g, respectively. Hot hardness measurements were obtained using a high-temperature hardness tester (Akashi AVK-CO) equipped with a heating and vacuum part (Akashi AVK-HF), using a load of 5 kg.

3. Results

3.1. Microstructure and hardness

Fig. 1 presents the SEM backscattered electron images (BEI) of Al00Ti05 (Fig. 1a and b) and Al02Ti05 (Fig. 1c and d). To obtain better contrast, these specimens were etched with dilute aqua regia to reveal the precipitates and grain boundaries. Al00Ti05 contains two phases, a matrix and some plate-like precipitates at the grain-boundaries of the matrix. In contrast, Al02Ti05 has an almost single-phase polycrystalline structure, except for a few isolated particles along the grain boundaries. These particles can only be observed in secondary electron images (SEI) of non-etched samples, as shown in the inset of Fig. 1d. Due to their small size relative to the EDS probe size, it is difficult to obtain their composition. We suspect that η formation is not completely suppressed by Al addition and these particles are the remnant η precipitates. The X-ray diffraction (XRD) patterns of the Ti05 alloys (i.e. Al00Ti05 and Al02Ti05) are composed of the same set of fcc peaks (Fig. 2). This indicates that the primary phase in these alloys is an fcc gamma phase, and the lattice constants are almost the same, 0.358 and 0.359 nm for Al00Ti05 and Al02Ti05, respectively. Peaks corresponding to the precipitates of Al00Ti05 were difficult to identify because they had weak intensity. The EDS analysis results in Table 1 show that the compositions of the matrix phases in Al00Ti05 and Al02Ti05 are similar to the overall compositions of Al00Ti05 and Al02Ti05. The precipitates at the grain boundary of Al00Ti05 contain more Ni, Co and Ti than the matrix, and the atomic ratio of (Ni + Co):Ti is close to 3:1. The morphology and composition of these

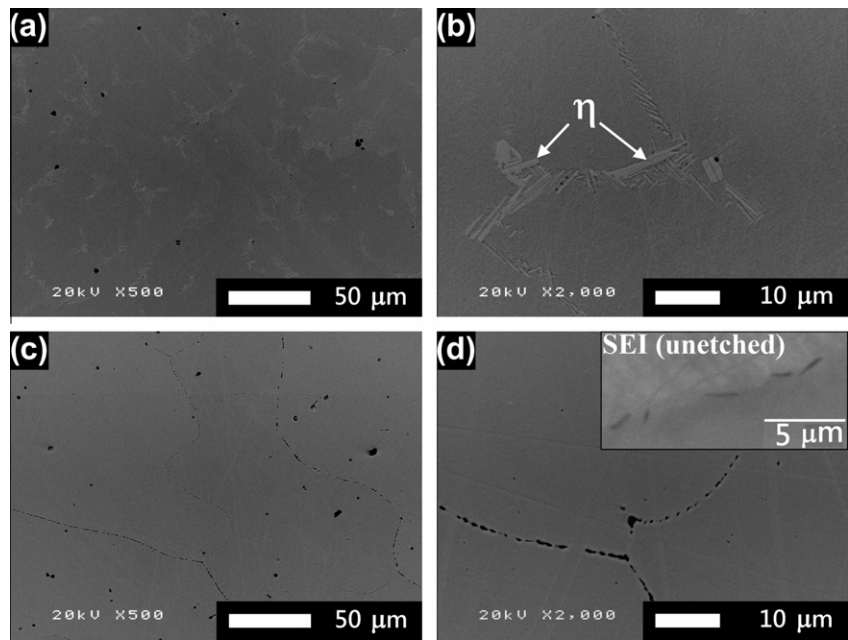


Fig. 1. SEM backscattering electron images of Ti05 alloys. (a and b) Al00Ti05; (c and d) Al02Ti05. These samples were etched with dilute aqua regia.

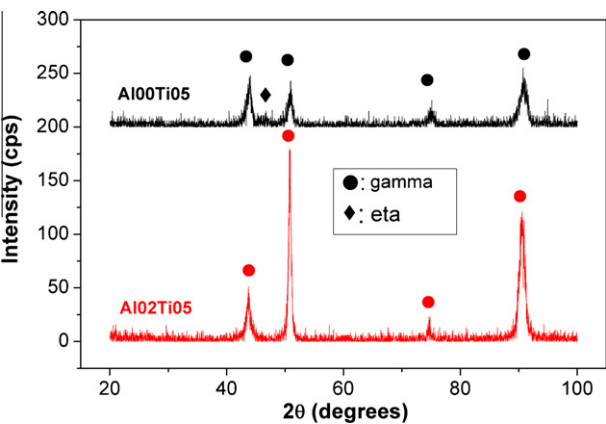


Fig. 2. XRD patterns of Ti05 alloys.

Table 1
Nominal and EDS-determined compositions (in at.%) of Al00Ti05 and Al02Ti05.

	Al00Ti05				Al02Ti05	
	Nominal	Overall	Matrix	η	Nominal	Overall
Al	—	—	—	—	3.5	3.8
Co	27.3	27.0	27.0	23.5	26.3	26.1
Cr	18.2	18.7	19.3	5.6	17.5	17.6
Fe	18.2	19.3	18.8	7.6	17.5	17.1
Ni	27.3	27.1	26.2	43.3	26.3	26.3
Ti	9.1	9.3	8.6	20.1	8.8	9.1

plate-shaped precipitates are very similar to those of the η -Ni₃Ti phase in (Ni, Ti)-rich alloy systems, which have a hexagonal close-packed (hcp) (D0₂₄) structure [22–25]. Since Ni and Co elements have similar atomic sizes and

chemical properties, they inter-substitute in multi-element alloys or superalloys [26–28]. Thus, in Al00Ti05, part of Ni in η -Ni₃Ti is likely to be replaced by Co to form the η -(Ni, Co)₃Ti phase. The morphology of the η -(Ni, Co)₃Ti phase is identical to η -Ni₃Ti and has been observed in many alloy systems that contain Ni, Co, and Ti [29–31]. Upon the addition of Al, the formation of η -(Ni, Co)₃Ti is suppressed, as seen in the Al02Ti05 alloy. This is because the hexagonal η phase has very little solubility for Al [32]. Hence, before the η phase can nucleate and grow, the diffusion of Al must occur so that the local Al content becomes lower than the solution limit of the η phase. This sets a kinetic constraint on the formation of the η phase; therefore, its formation is suppressed.

When the molar ratio of Ti is increased from 0.5 to 1.0, the microstructure becomes more complex, as seen from the SEM images of the Ti10 alloys (i.e. Al00Ti10 and Al02Ti10) in Fig. 3. Both alloys have a dendrite–interdendrite structure. The dendrite (DR) regions of the two alloys have very similar compositions and are Ti-deficient (Tables 2 and 3). In contrast, the interdendrite (ID) regions of both alloys contain different phases. The ID regions of Al00Ti10 are composed of a (Co, Cr, Fe)-rich matrix and a high volume fraction of white blocky precipitates, 5–10 μ m in size. EDS analysis shows that the precipitates have a composition similar to that of the η -(Ni, Co)₃Ti phase in Al00Ti05. Meanwhile, the peaks corresponding to the η -(Ni, Co)₃Ti phase (D0₂₄, hcp) are also observed in the XRD pattern of Al00Ti10 (Fig. 4), with the lattice constants $a = 0.509$ nm and $c = 0.85$ nm. We therefore conclude that these white blocky precipitates are coarse η -(Ni, Co)₃Ti phases with hcp (D0₂₄) structures. Some minor peaks in the XRD pattern could not be identified (e.g. $\sim 79^\circ$ in

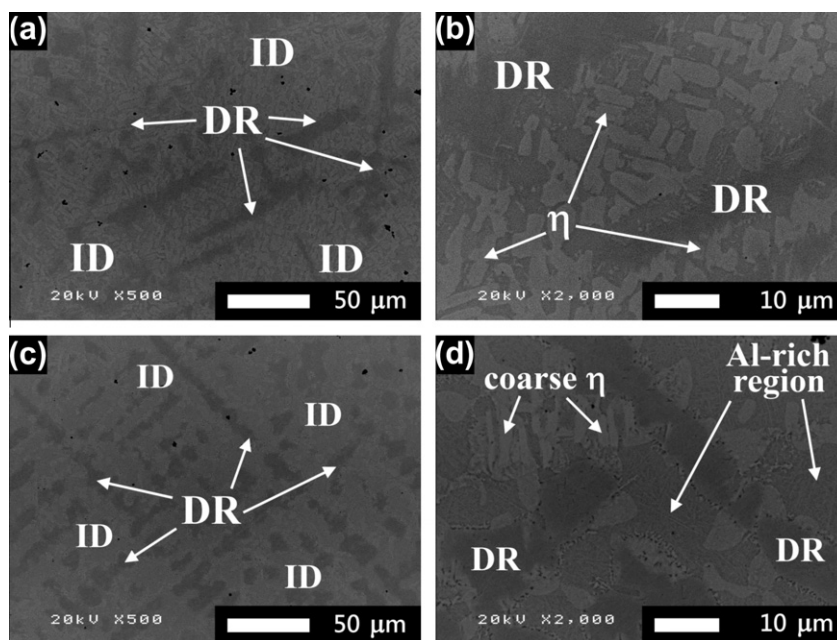


Fig. 3. SEM backscattering electron images of Ti10 alloys. (a and b) Al00Ti10; (c and d) Al02Ti10.

Table 2
Nominal and EDS-determined compositions (in at.%) of Al00Ti10.

	Nominal	Overall	DR	ID	
				Matrix	Coarse η
Co	25.0	24.5	24.9	25.5	24.4
Cr	16.7	17.1	20.1	24.1	5.8
Fe	16.7	16.3	20.9	20.3	8.7
Ni	25.0	24.6	23.0	14.5	39.3
Ti	16.7	17.5	11.1	15.6	21.8

Al00Ti10 and Al02Ti10) because they had low intensity and were limited in number. It is conjectured that these weak peaks might be contributed by some unknown minor phases.

The structure in the ID regions of Al02Ti10 is more complex. The amount of coarse η -(Ni, Co)₃Ti in the ID of Al02Ti10 is evidently reduced, and new regions with high Al content emerge. Further observation shows that these Al-rich regions contain a dense distribution of needle-like precipitates embedded in a matrix. To investigate the composition and morphology of these needles, the matrix was dissolved by dilute aqua regia to leave only the needles. From the SEM image shown in Fig. 5, it can

be seen that the needles grow along a certain orientation, forming a typical Widmanstätten structure. The composition of these needles is very similar to the η precipitates in Al00Ti05 and Al00Ti10, and it belongs to the η -(Ni, Co)₃Ti phase. Therefore, these are the Widmanstätten η , or needle-like η , that can be found in many Ni-base superalloys [23,29,33,34]. Comparing the overall composition of the Al-rich region and the composition of needle-like η , one can see that the matrix in the Al-rich region is Al-, Cr-, Fe-rich and Ni-deficient compared to the needles. This means that its composition is closer to the fcc phase; and therefore it is conjectured that the matrix also has an fcc structure.

Comparing the microstructures and XRD patterns of Ti05 and Ti10 alloys shows that increasing the Ti molar ratio from 0.5 to 1 significantly increases the quantity of the η -(Ni, Co)₃Ti precipitates. Since Ti has a large negative enthalpy of mixing with both Ni and Co (−35 and −28 kJ mol^{−1}, respectively), when the content of Ti in an Ni-, Co-rich environment is higher than the solution limit, the (Ni, Co)₃Ti phase precipitates, as frequently observed in superalloys [30,32,35]. The solution limit of Ti is found to be around 11 at.%. This can be seen by comparing the

Table 3
Nominal and EDS-determined compositions (in at.%) of Al02Ti10.

	Nominal	Overall	DR	ID			
				Matrix	Coarse η	Al-rich region (overall)	Al-rich region (needles only)
Al	3.2	3.9	1.7	0.7	1.4	5.0	1.2
Co	24.2	23.9	23.8	23.6	22.7	24.3	23.2
Cr	16.1	15.6	19.2	26.2	5.5	9.7	5.2
Fe	16.1	15.6	19.6	22.1	8.5	11.5	8.0
Ni	24.2	23.6	24.6	12.2	40.3	27.2	38.2
Ti	16.1	17.4	11.2	15.3	21.5	22.3	24.3

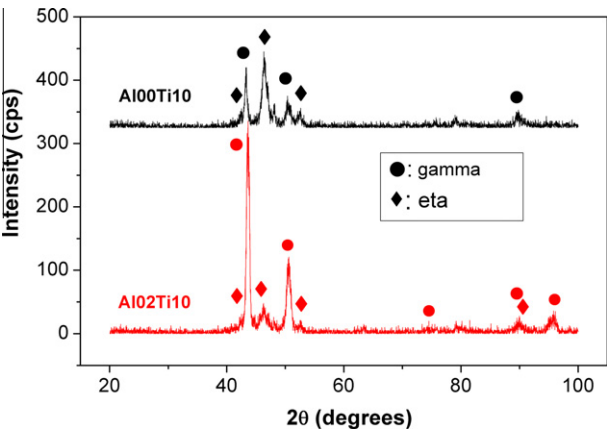


Fig. 4. XRD patterns of Ti10 alloys.

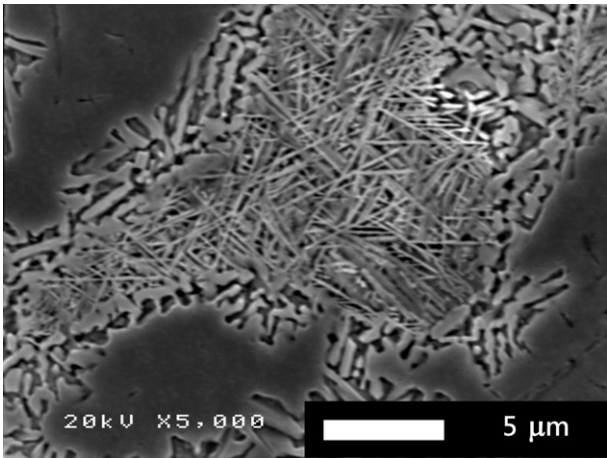


Fig. 5. SEM secondary electron image of the η phase with a needle-like Widmanstätten structure in the ID of Al02Ti10. The sample was etched with dilute aqua regia.

composition of the matrix in Al00Ti05 and the DR of Al00Ti10. Although the Ti content in Al00Ti10 is doubled, the Ti concentration in the DR of Al00Ti10 only increased from 8.6 to 11.1 at.%. Comparing Al02Ti05 and Al02Ti10 also gives us the same value.

The XRD patterns of the two Ti10 alloys also reveal that the addition of Al still effectively suppresses the formation of η . Indeed, in the Al00Ti10, coarse η precipitates are distributed over the whole ID region. In the Al02Ti10 alloy, however, coarse η is only observed in certain ID areas, where Al has been substantially removed. In most ID regions, only local, short-range redistributions of Al can take place; therefore, needle-like precipitates are prevalent instead of coarse ones.

The macrohardness values of the four HEAs and the microhardness values of selected areas are listed in Table 4. Alloys with lower Ti content (i.e. Al00Ti05 and Al02Ti05) have lower hardness because they are composed mainly of the fcc gamma phase. The slightly higher hardness of Al00Ti05 is due to the η precipitates at the grain boundaries. When the molar ratio of Ti is increased to 1.0, the hardness of the alloys increases. The overall hardness of Al00Ti10 is about HV 650, higher than Al00Ti05 and Al02Ti05. This is because a large amount of η phase forms in the ID regions of Al00Ti10, which raises the microhardness of the ID region to HV 900. Although the amount of coarse η phases in the ID regions of Al02Ti10 is less than that in the ID regions of Al00Ti10, the needle-like η phase with Widmanstätten structures provides an even higher hardness (up to HV 1200). Therefore, Al02Ti10 has the highest overall hardness (HV 717).

3.2. Wear test results

Fig. 6 compares the hardness and wear resistance of the six alloys tested in this research. HEAs with higher hardness (Al00Ti10 and Al02Ti10) have better wear resistance than the softer alloys (Al00Ti05 and Al02Ti05). Al02Ti10 has the highest wear resistance among the four HEAs, which is about twice that of Al00Ti10. It is worthwhile to note that the hardness of Al02Ti10 is close to that of SUJ2 and apparently lower than SKH51, but the wear resistance of Al02Ti10 is significantly higher than these two reference conventional alloys (3.6 and 2 times higher than those of SUJ2 and SKH51, respectively). This indicates that Al02Ti10 has outstanding anti-wear properties.

Fig. 7a and c shows the secondary electron images (SEI) of the wear surface of the Al00Ti05 and Al02Ti05 specimens. The wear surface morphologies of these two alloys are similar, showing noticeable plastic deformation features and grooves. EDS analysis (not shown) reveals that most of the wear surface contains no oxygen except the dark regions in the BEI (Fig. 7b and d). The wear debris is flake-like (Fig. 9a and b) and also contains no oxygen (Table 5), inferring that most debris particles are worn off from Al00Ti05 and Al02Ti05 without significant oxidation.

In sharp contrast to the wear surfaces of Ti05 alloys, only shallow scratches and some significant oxidation features are observed on the wear surfaces of Al00Ti10 (Fig. 8a and b) and Al02Ti10 (Fig. 8c and d), and the dendrite–interdendrite structures can still be recognized. EDS analysis (not shown) on the DR and ID areas show apparent signals of oxygen, but their compositions are quite different. The composition of the worn ID is close to that of

Table 4
Overall-hardness and microhardness of each phase in the four HEAs.

	Al00Ti05 overall	Al02Ti05 overall	Al00Ti10			Al02Ti10			
			Overall	ID	DR	Overall	ID (Al-rich region)	ID (coarse η)	DR
Hardness (HV)	509 ± 11	487 ± 5	654 ± 7	924 ± 33	617 ± 13	717 ± 13	1197 ± 110	910 ± 24	619 ± 12

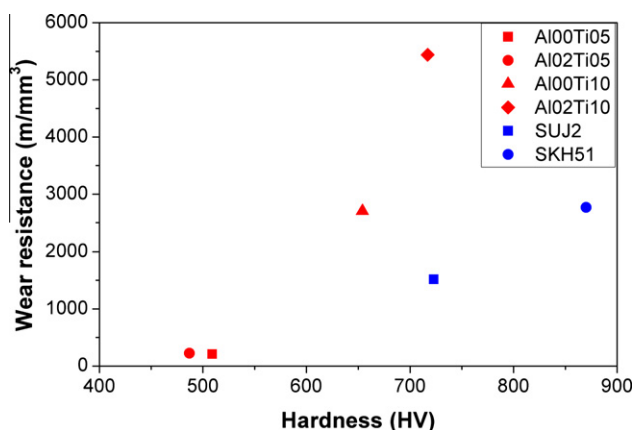


Fig. 6. Pin-on-disk adhesive wear resistance vs. hardness of Al00Ti05, Al02Ti05, Al00Ti10, Al02Ti10, SUJ2 and SKH51.

the alloy except for the presence of oxygen. The DR regions, due to their lower hardness, are severely worn off and powdery black particles accumulate in these worn-off dents. Such black particles are Fe-rich oxides according to the EDS analysis results (Table 6). Their composition is similar to that of the wear debris from the Ti10 alloys (Table 5). Due to the higher Fe concentration of the wear debris and the accumulated black particles, it is highly probable that most of them originate from the wear of the SKH51 disk.

3.3. Oxidation weight gain test and hot hardness measurement

As previously mentioned, some significant oxidation features are observed on the sample surfaces after the wear test. This means that during the dry sliding wear process

the temperature of the wear surface increases substantially due to frictional heating. Indeed, our estimation of the surface flash temperature according to Ashby et al.'s research [36] suggests that during our sliding wear test conditions the flash temperature of the local contact areas on Al00Ti10 and Al02Ti10 may be higher than 800 °C. For SUJ2, which has a higher thermal conductivity, the flash temperature can also be higher than 600 °C. At these temperatures, both oxidation and softening become significant enough to influence the wear properties of the alloys. To understand the effects of high-temperature oxidation and softening, oxidation weight gain tests and hot hardness measurements were carried out.

Fig. 10 shows the oxidation weight gain of SUJ2, SKH51 and HEAs after being heat-treated at 600 and 800 °C for 24 h in air. The weight gains of the HEAs are much smaller than those of SUJ2 and SKH51 at both 600 and 800 °C. The hot hardness measurements of SUJ2, SKH51, Al00Ti10 and Al02Ti10 as a function of temperature are shown in Fig. 11. The hardness of Al00Ti10 and Al02Ti10 alloys is higher than SKH51 at 700 °C and above. Both the oxidation weight gain test and the hot hardness measurement results indicate that HEAs have better thermal stability than SUJ2 and SKH51 steels. Further discussions on the oxidation resistance and hot hardness and their effects on the wear behavior are included in Section 4.2.

4. Discussion

4.1. Wear mechanism

The four HEAs in this study are categorized into two groups based on their Ti content. These two groups have

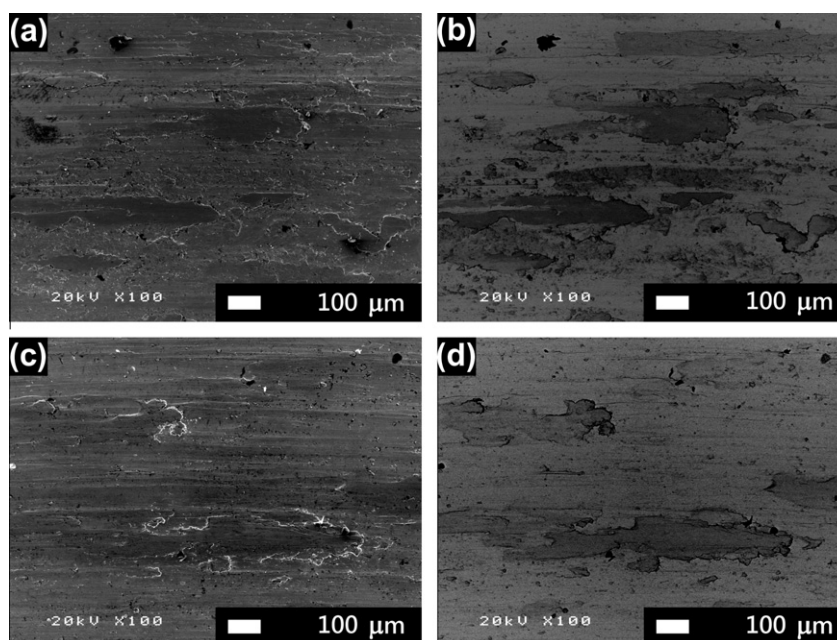


Fig. 7. SEM images of the wear surface of Al00Ti05: (a) SEI, (b) BEI; and Al02Ti05: (c) SEI, (d) BEI.

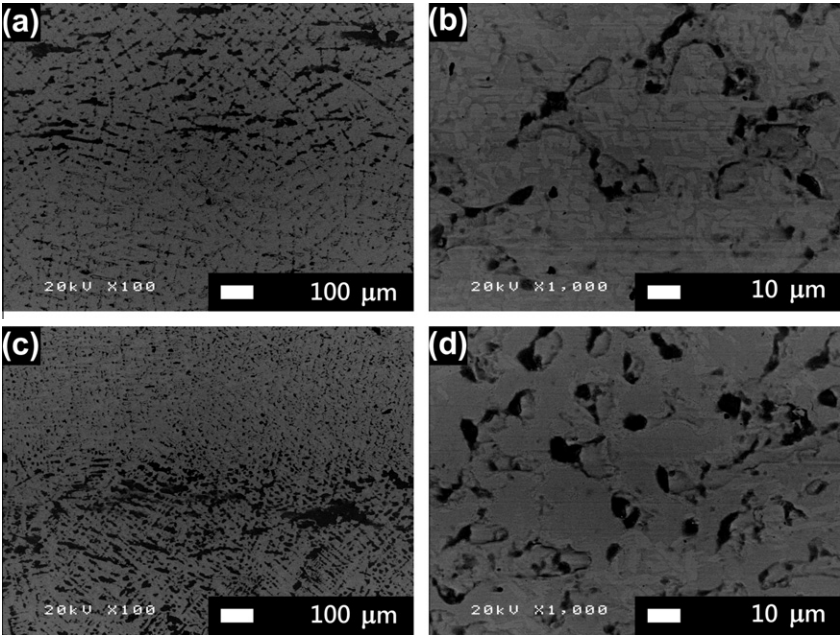


Fig. 8. SEM BEI of the wear surface on Al00Ti10: (a) magnification 100×, (b) magnification 1000×; and Al02Ti10: (c) magnification 100×, (d) magnification 1000×.

Table 5
Composition (in at.%) of the wear debris by EDS analysis.

	O	Al	Ti	Cr	Fe	Co	Ni
Al00Ti05	–	–	9.3	18.5	18.7	27.3	26.3
Al02Ti05	–	3.8	8.6	18.0	17.8	26.4	25.3
Al00Ti10	68.1	–	3.3	3.6	14.6	4.8	5.0
Al02Ti10	69.4	0.8	2.7	3.0	17.8	3.9	4.4

Table 6
Composition (in at.%) of the dark areas on the wear surface of Al00Ti10 and Al02Ti10 by EDS analysis.

	O	Al	Ti	Cr	Fe	Co	Ni
Al00Ti10	66.4	–	3.4	4.2	17.3	4.5	4.3
Al02Ti10	63.3	1.0	4.0	5.1	17.6	4.6	4.5

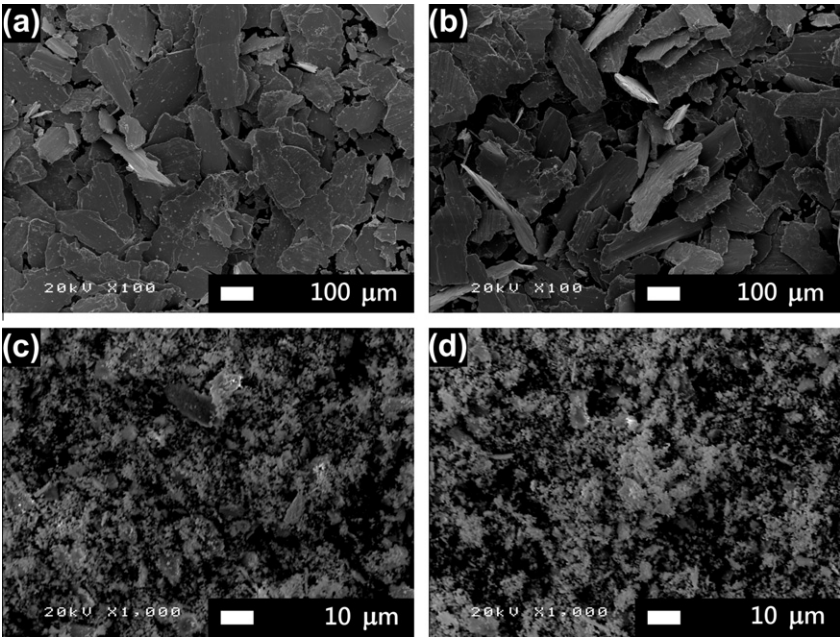


Fig. 9. SEM microstructure of wear debris of (a) Al00Ti05; (b) Al02Ti05; (c) Al00Ti10; (d) Al02Ti10.

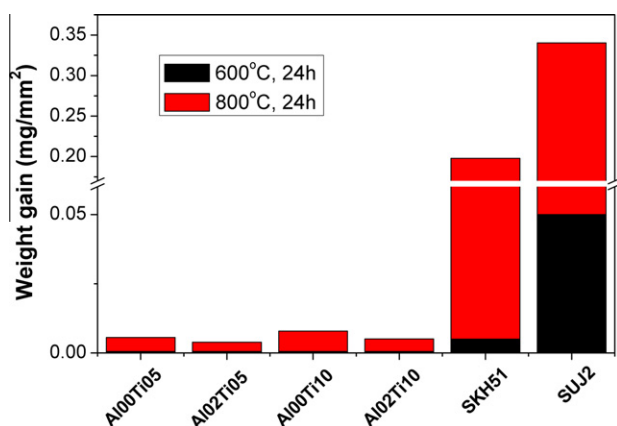


Fig. 10. Weight gain test of Al00Ti05, Al02Ti05, Al00Ti10, Al02Ti10, SUJ2 and SKH51 specimens at 600 and 800 °C for 24 h. The data is presented as weight increase per unit area after test.

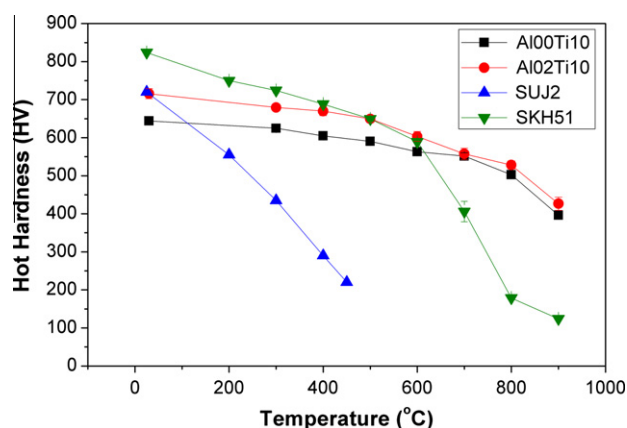


Fig. 11. Hot hardness vs. temperature plots for Al00Ti10, Al02Ti10, SUJ2 and SKH51 specimens from room temperature to 900 °C.

different hardnesses and microstructures, and therefore give rise to two different wear mechanisms against the SKH51 disk. The wear mechanism of the Al00Ti05 and Al02Ti05 alloys is delamination wear (i.e. severe plastic deformation) [37–40]. This can be confirmed by the obvious scratches on the wear surface and the plate-like wear debris whose composition is similar to the alloys. The dominance of delamination wear is due to the significantly higher hardness of the wear plate (SKH51) relative to Ti05 alloys, which are composed mainly of a soft fcc phase with a hardness of $\sim$ HV 500. When slid against the SKH51 disk (whose hardness is HV 890), the Ti05 alloys are plastically deformed and scratched. Further, when the degree of plastic deformation exceeds a certain threshold, the alloys begin to form cracks and split from the surface, forming the metallic plate-like wear debris.

In contrast, the wear mechanism of the Al00Ti10 and Al02Ti10 alloys is mild oxidative wear [40–42]. This can be judged from two reasons. The first is the shallow wear surface scratches, the significant oxidation feature on the wear surface, and the oxide-based wear debris.

The second is the apparently higher Fe content in the debris relative to that of other elements, which indicates that the SKH51 disk is seriously worn. Although the overall hardness of Al00Ti10 and Al02Ti10 is lower than the SKH51 disk, the ID microhardness is over HV 900, which is higher than the SKH51 disk. Therefore, the Ti10 alloys are strong enough to resist the wear against the disk, and the alloy surface will not be worn off quickly during the wear test. In fact, they are able to withstand being worn for a long time and are therefore continuously heated due to friction. Such frictional heating raises the temperature of the local contact areas on the surface, leading to surface oxidation. Indeed, as mentioned previously, oxidized layers were observed on worn ID regions, and their compositions were similar to that of the alloy except for the presence of oxygen. This oxidized layer protects the specimen from direct metal–metal contact and reduces the degree of adhesive wear [41–43]; thus, the wear rate of the specimen is further decreased.

4.2. Mechanism of the outstanding wear resistance in Al00Ti10 and Al02Ti10

Compared with the conventional wear-resistant steels SKH51 and SUJ2, Al00Ti10 and Al02Ti10 exhibit even higher resistance to adhesive wear. The wear resistance of SKH51 and SUJ2 originates from their hard carbide phases [44–46]. Similarly, the hard η phase in Ti10 alloys should also be an important factor in their anti-wear properties. However, the carbides in SKH51 and SUJ2 are harder than the η phases in Ti10 alloys [45]. Therefore, some other factors must also play important roles in the wear of Ti10 alloys. The higher wear resistances of Al00Ti10 and Al02Ti10 are attributed to their better oxidation resistances and superior hot hardness values.

Oxidation resistance plays a critical role in the mild oxidative wear mechanism. In this mechanism, the surface oxide layer prevents direct metal–metal contact and adhesive wear. Thus, the volume loss comes mainly from the wear of the oxide layer [47,48]. It has been shown that delamination of the oxide layer takes place due to internal stress when it reaches a critical thickness [42]. Therefore, in mild oxidative wear, the better the oxidation resistance of the alloy (i.e. forming thin, dense and well-attached oxide layers), the less the wear loss.

As seen from the weight gains at 600 °C (Fig. 10), the HEAs are almost not oxidized, but SKH51 and SUJ2 both show apparent weight gain. At 800 °C, unambiguous oxidation signals are observed in HEAs. However, the weight gains of SKH51 and SUJ2 at 800 °C are 26 and 39 times higher, respectively, than those of the HEAs. This is especially so for SUJ2, whose oxide layer peels off from the specimen surface after heating for 24 h at 600 and 800 °C. These results indicate that the surfaces of the steels may oxidize rapidly under the wear test condition of the present research. However, the HEAs have very good oxidation resistances at elevated temperature, and the dense

and well-adhering oxide layers formed during the wear test can protect the underlying materials from wear loss.

Another reason for the superior wear resistance of Al00Ti10 and Al02Ti10 is their high hot hardness values. Under the oxidational wear mechanism, the wear mainly occurs at the real contact areas on the surfaces. As previously mentioned, the temperature of these local areas (i.e. flash temperature) could be as high as 800 °C because of frictional heating, and the temperature of adjacent areas can also increase to 500–600 °C. Such high temperatures can soften the alloys considerably, resulting in a decrease in wear resistance. The hot hardness values of SUJ2, SKH51, Al00Ti10 and Al02Ti10 as a function of temperature are shown in Fig. 11. It can be seen that although SUJ2 and SKH51 have very high hardness values at room temperature, their hardness decreases rapidly as the temperature increases. For example, the hardness of SUJ2 decreases from HV 723 to less than HV 250 at 450 °C. SKH51 has a better resistance to thermal softening, but its hardness still decreases rapidly above 600 °C. In contrast, the softening rate of Al00Ti10 and Al02Ti10 is apparently slower. In fact, both Al00Ti10 and Al02Ti10 have hardness values higher than HV 500 even at 800 °C. This implies that at 800 °C HEAs are at least 2.5 times harder than SKH51! Such good resistance to softening in HEAs has also been reported in another HEA system [16], and originates from the enhanced solution hardening effect in multi-principle-element solutions. The wear surfaces of Al00Ti10 and Al02Ti10 will be apparently stronger than SUJ2 and SKH51 during the wear test, and their better wear resistance is thus reasonable.

In this work, the Al02Ti10 alloy, which has high hardness, superior oxidation resistance and remarkable hot hardness, shows the highest resistance to adhesive wear among all the tested materials. Under mild oxidational wear mechanism, the well-adhering surface oxide layer on Al02Ti10 protects the underlying material. The better resistance to thermal softening implies that the local contact areas will have a higher hardness against adhesive wear. The superior wear resistance of this alloy thus has potential application in wear parts, especially at elevated temperatures.

5. Conclusions

1. Four HEAs designed in this study are categorized into two groups based on their Ti content. The main phase in Ti05 alloys (Al00Ti05 and Al02Ti05) is a gamma phase with an fcc structure. A small amount of plate-like η phase precipitate is observed at the grain boundaries of Al00Ti05. η precipitation is suppressed in Al02Ti05 due to the addition of Al. The hardness values of Al00Ti05 and Al02Ti05 are HV 509 and HV 487, respectively.
2. Increasing the molar ratio of Ti from 0.5 to 1.0 induces a dendritic structure in the Ti10 alloys (Al00Ti10 and

Al02Ti10) and significantly increases the volume fraction of η precipitates. A large amount of coarse η is observed in the ID region of Al00Ti10. The addition of Al in Al02Ti10 decreases the amount of the coarse η phase in the ID regions and triggers the formation of the needle-like η phase with Widmanstätten structures in the Al-rich regions of the ID. This occurs because of the difficulty in effectively relocating Al. The hardness values of Al00Ti10 and Al02Ti10 become HV 654 and HV 717, respectively, because of the strengthening effect from the hard η phase.

3. Due to the lower hardness of Al00Ti05 and Al02Ti05 compared to the wear disk, they undergo severe delamination wear. Al00Ti10 and Al02Ti10 alloys have a higher hardness and are much more resistant to sliding wear. The shallow scratches and oxidation features on the wear surface and the Fe-rich wear debris of Al00Ti10 and Al02Ti10 show that their wear behavior belongs to mild oxidational wear.
4. The wear resistance of Al00Ti10 and Al02Ti10 are markedly higher than that of the two reference conventional steels. The hardness of Al02Ti10 is similar to that of SUJ2, but the wear resistance of the former is 3.6 times that of the latter. The wear resistance of Al02Ti10 is twice that of SKH51, but the hardness of the latter is a lot higher. The superior wear resistances of Al00Ti10 and Al02Ti10 relative to that of conventional steels originate from their higher oxidation resistance and lower high-temperature softening rates.

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References

- [1] Huang KH. MS thesis, Department of Materials Science and Engineering, NTHU, Taiwan; 1996.
- [2] Yeh JW, Chen SK, Lin SJ, Gan JY, Chin TS, Shun TT, et al. *Adv Eng Mater* 2004;6:299.
- [3] Yeh JW, Chen YL, Lin SJ, Chen SK. *Mater Sci Forum* 2007;560:1.
- [4] Ranganathan S. *Curr Sci* 2003;85:1404.
- [5] Yeh JW. *Ann Chim – Sci Mater* 2006;31:633.
- [6] Tong CJ, Chen YL, Chen SK, Yeh JW, Shun TT, Tsau CH, et al. *Metall Mater Trans A* 2005;36A:881.
- [7] Hsu CY, Yeh JW, Chen SK, Shun TT. *Metall Mater Trans A* 2004;35A:1465.
- [8] Huang PK, Yeh JW, Shun TT, Chen SK. *Adv Eng Mater* 2004;6:74.
- [9] Tong CJ, Chen MR, Chen SK, Yeh JW, Shun TT, Lin SJ, et al. *Metall Mater Trans A* 2005;36A:1263.
- [10] Chen MR, Lin SJ, Yeh JW, Chen SK, Huang YS, Chuang MH. *Metall Mater Trans A* 2006;37A:1363.
- [11] Chen MR, Lin SJ, Yeh JW, Chen SK, Huang YS, Tu CP. *Mater Trans* 2006;47:1395.
- [12] Wu JM, Lin SJ, Yeh JW, Chen SK, Huang YS. *Wear* 2006;261:513.
- [13] Zhou YJ, Zhang Y, Wang YL, Chen GL. *Appl Phys Lett* 2007;90:181904.

- [14] Wang XF, Zhang Y, Qiao Y, Chen GL. *Intermetallics* 2007;15:357.
- [15] Wang YP, Li BS, Ren MX, Yang C, Fu HZ. *Mater Sci Eng A* 2008;491:154.
- [16] Hsu CY, Wang WR, Tang WY, Chen SK, Yeh JW. *Adv Eng Mater* 2010;12:44.
- [17] Zhu JM, Zhang HF, Fu HM, Wang AM, Li H, Hu ZQ. *J Alloys Compd* 2010;497:52.
- [18] Hsu CY, Sheu TS, Yeh JW, Chen SK. *Wear* 2010;268:653.
- [19] Singh S, Wanderka N, Murty BS, Glatzel U, Banhart J. *Acta Mater* 2011;59:182.
- [20] Tung CC, Yeh JW, Shun TT, Chen SK, Huang YS, Chen HC. *Mater Lett* 2007;61:1.
- [21] Chen ST, Tang WY, Kuo YF, Chen SY, Tsau CH, Shun TT, et al. *Mater Sci Eng A* 2010;527:5818.
- [22] Davis JR, editor. *ASM specialty handbook: nickel, cobalt, and their alloys*. Materials Park, OH: ASM International; 2000. p. 68.
- [23] Davis JR, editor. *ASM specialty handbook: nickel, cobalt, and their alloys*. Materials Park, OH: ASM International; 2000. p. 293.
- [24] Heo Y-U, Takeguchi M, Furuya K, Lee H-C. *Acta Mater* 2009;57:1176.
- [25] Schnitzer R, Schober M, Zinner S, Leitner H. *Acta Mater* 2010;58:3733.
- [26] Saito Y, Harada H. *Mater Sci Eng A* 1997;223:1.
- [27] He Y, Liu H, Shin K, Lee CG. *J Alloys Compd* 2008;463:546.
- [28] Kitashima T, Yokokawa T, Yeh AC, Harada H. *Intermetallics* 2008;16:779.
- [29] Zhao J, Ravikumar V, Beltran A. *Metall Mater Trans A* 2001;32:1271.
- [30] Cui CY, Gu YF, Ping DH, Harada H, Fukuda T. *Mater Sci Eng A* 2008;485:651.
- [31] Gui C, Sato A, Gu Y, Harada H. *Metall Mater Trans A* 2005;36:2921.
- [32] Muzyka DR. In: Sims CT, Hagel WC, editors. *The superalloys*. New York: Wiley-Interscience; 1972. p. 113.
- [33] Asgari S. *Metall Mater Trans A* 2006;37:2051.
- [34] Li X, Zhang J, Rong L, Li Y. *Mater Sci Eng A* 2008;488:547.
- [35] Cui C, Sato A, Gu Y, Ping D, Harada H. *Metall Mater Trans A* 2006;37:3183.
- [36] Ashby MF, Abulawi J, Kong HS. *Tribol Trans* 1991;34:577.
- [37] Suh NP. *Wear* 1973;25:111.
- [38] Suh NP. *Wear* 1977;44:1.
- [39] Rigney DA, Chen LH, Naylor MGS, Rosenfield AR. *Wear* 1984;100:195.
- [40] Lim SC, Ashby MF. *Acta Metall* 1987;35:1.
- [41] Quinn TFJ. *Tribol Int* 1983;16:257.
- [42] Stott FH. *Tribol Int* 1998;31:61.
- [43] Lim SC. *Tribol Int* 2002;35:717.
- [44] Söderberg S, Hogmark S. *Wear* 1986;110:315.
- [45] Vardavoulias M. *Wear* 1994;173:105.
- [46] Serna MM, Rossi JL. *Mater Lett* 2009;63:691.
- [47] Quinn TFJ. *Br J Appl Phys* 1962;13:33.
- [48] Quinn TFJ. *Wear* 1971;18:413.